

Q3

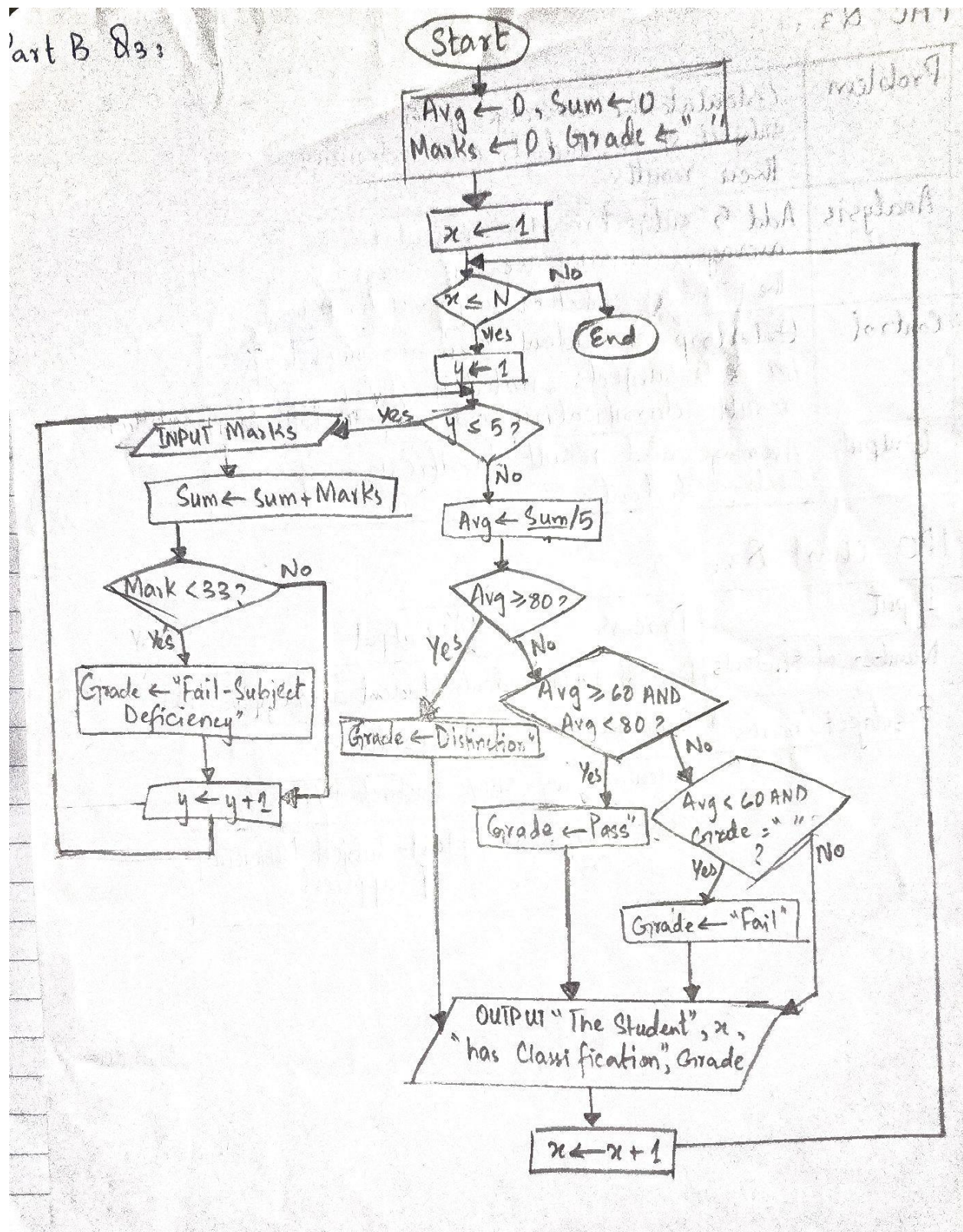
Pseudocode:

```

START
DECLARE Avg; Marks : FLOAT
DECLARE Sum : FLOAT
DECLARE Grade; STRING
Avg ← 0, Sum ← 0, Marks ← 0
Grade ← " "
FOR x ← 1 to N
    Sum ← 0
    FOR y ← 1 to 5
        Marks ← INPUT("Enter Marks for subject", y)
        Sum ← Sum + Marks
        IF Mark < 33 THEN
            Grade ← "Fail - Subject Deficiency"
        EndIF
    NEXT y
    Avg ← Sum Sum / 5
    IF Avg ≥ 80 THEN
        Grade ← "Distinction"
    ELSE IF Avg ≥ 60 THEN OR AND AND Avg < 80 THEN
        Grade ← "Pass"
    ELSE IF Avg < 60 AND Grade = " " THEN
        Grade ← "Fail"
    EndIF
    NEXT OUTPUT "The Student", x, "has Average"
    OUTPUT "The Student", x, "has average C"
    OUTPUT "The Student", x, "has Classification:", Grade
    NEXT Sum ← 0
END NEXT x

```

Flowchart:



IPO and PAC chart:

PAC Q3.

Problem	Calculate the average of 5 subject of N students and determine their results.
Analysis	Add 5 subject marks, calculate average, classify average accordingly then check whether any mark is below 33
Control	Outer loop for students then and inner loop for 5 subjects, nested if statements for result classification and subject deficiency.
Output	Average and result classification of each student.

IPO chart Q3.

Input	Process	Output
Number of Students	Repeat for N students	Student average
5 subject marks	Sum marks	Result
	Calculate $avg = \frac{sum}{5}$	Distinction/Pass/Fail
	Check if any mark < 33	Fail-Subject Deficiency if applied.